AP CALCULUS AB - UNIT 1

1.5 Determining Limits Using Algebraic Properties of Limits

Lesson Overview

- Apply constant, sum, difference, product, quotient, power, root, and composition laws.
- Check every hypothesis before using a limit law.
- Separate known limit values from function values at the target.
- Justify direct substitution through continuity and domain conditions.

Key Knowledge and Vocabulary

Linearity. $\lim(af + bg) = a \lim f + b \lim g$ when the component limits exist.

Products and powers. $\lim(fg) = (\lim f)(\lim g)$ and $\lim[f(x)]^n = (\lim f)^n$.

Quotient condition. $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ only when the denominator limit is nonzero.

Even roots. For real-valued limits, the landing value must lie in the appropriate domain; endpoint cases may be one-sided.

Composition. If $g(x) \to L$ and the outer function is continuous at L, then $F(g(x)) \to F(L)$.

Explanation and Strategy

Step 1. Write the limit of each changing component before combining them.
Step 2. A failed hypothesis means a law is inconclusive, not that the limit automatically fails.
Step 3. Direct substitution is a theorem-based shortcut for continuous functions at allowed inputs.

Solved Examples

Worked Example: Linear combination

Given $\lim_{x\to a} f(x) = 5$ and $\lim_{x\to a} g(x) = -2$, find $\lim_{x\to a}[3f(x) - 4g(x) + 1]$.

Solution. $3(5) - 4(-2) + 1 = 24$.

Worked Example: Nested structure

If $f(x) \to 3$ and $g(x) \to -1$, evaluate $\lim \sqrt{2f(x)^2 - g(x)}$.

Solution. The inside approaches $2(3)^2 - (-1) = 19 > 0$. The square-root function is continuous there, so the limit is $\sqrt{19}$.

Worked Example: Failed quotient hypothesis

If $f(x) \to 0$ and $g(x) \to 0$, what does the quotient law say about $\lim f(x)/g(x)$?

Solution. Nothing conclusive. The denominator-limit hypothesis fails. The actual quotient limit might be finite, infinite, or nonexistent depending on the functions.

Leveled Practice

Shared Representation / Data

	$\lim_{x\to 2} f(x)$	$\lim_{x\to 2} g(x)$	$\lim_{x\to 2} h(x)$
Value	4	−3	0

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Using the shared limit data, find $\lim_{x\to 2}[f(x) + g(x)]$.

2. Using the shared limit data, find $\lim_{x\to 2}[2f(x) - 5g(x)]$.

3. Using the shared limit data, find $\lim_{x\to 2} f(x)g(x)$.

4. Using the shared limit data, find $\lim_{x\to 2}[g(x)]^2$.

Medium Level

Apply the idea in one or two connected steps.

5. Using the shared data, evaluate $\lim_{x\to 2} \sqrt{f(x) + 5}$.

6. Using the shared data, evaluate $\lim_{x\to 2} \frac{f(x)-1}{g(x)+5}$.

7. If $\lim_{x\to c} u(x) = -2$, evaluate $\lim_{x\to c}[u(x)^3 + 4u(x) - 7]$.

8. Evaluate $\lim_{x\to 1} \frac{x^2+3x+1}{2x+5}$ by direct substitution and state why it is valid.

Hard Level

Connect representations, conditions, and justification.

9. If $f(x) \to 4$ and $g(x) \to 0$, can the quotient law evaluate $\lim f(x)/g(x)$? Explain precisely.

10. Given $u(x) \to 9$ and $u(x) \geq 0$ near the target, evaluate $\lim \frac{\sqrt{u(x)}-1}{\sqrt{u(x)}+1}$.

11. If $g(x) \to 2^-$ and $F(t) = \begin{cases} t^2, & t < 2 \\ 7, & t \geq 2 \end{cases}$, find $\lim F(g(x))$.

12. If $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c}[f(x) + g(x)] = M$, determine $\lim_{x \to c} g(x)$.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Assume $f(x) \to -4$. Determine whether $\lim \sqrt{f(x) + 4}$ is automatically valid as a real two-sided limit.

14. If $f(x) \to 2$ and $f(x) > 2$ near the target, evaluate $\lim \frac{|f(x)-2|}{f(x)-2}$.

15. Suppose $f(x)g(x) \to 12$ and $f(x) \to 3$. Prove that $g(x) \to 4$.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Let $f(x) \to a$ and $g(x) \to b$. Find a condition on a, b that guarantees $\lim \frac{f(x)^2 - g(x)^2}{f(x) - g(x)} = a + b$ using only limit laws.

17. Assume $f(x) \to L$ and $f(x)^2 \to 9$. What values of L are possible? What extra condition forces $L = 3$?

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. If $\lim f = 2$ and $\lim g = -5$, then $\lim(3f - g^2)$ equals

(A) -19 **(B)** -7
(C) 19 **(D)** 31

2. Which condition is required to use the quotient law for $\lim f(x)/g(x)$?

(A) $f(c)$ exists **(B)** $g(c) \neq 0$
(C) $\lim g(x) \neq 0$ **(D)** f and g are differentiable

3. Why is direct substitution valid for $\lim_{x \to 3}(x^3 - 2x + 4)$?

(A) The expression is a polynomial and is continuous at 3. **(B)** All limits equal function values.
(C) The denominator is zero. **(D)** The graph has a hole.

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator gives $u(x) \to 1.73205$ and $v(x) \to -0.50000$. Estimate $\lim[u(x)^2 + 4v(x)]$.

(A) -1 **(B)** 0

(C) 1 **(D)** 5

5. A graphing calculator suggests $g(x) \to 0$ while $f(x) \to 5$ near $x = 2$. Which calculation is not licensed by the basic limit laws?

(A) $\lim(f + g)$ **(B)** $\lim(fg)$

(C) $\lim f^2$ **(D)** $\lim(f/g)$

Free-Response Practice – No Calculator

At $x = a$, suppose $\lim f(x) = 3$, $\lim g(x) = -2$, and $\lim h(x) = 4$.

(a) Find $\lim[2f(x)g(x) - h(x)]$. [2 points]

(b) Find $\lim \sqrt{f(x)^2 + h(x)}$ and justify the root step. [2 points]

(c) Can the quotient law determine $\lim \frac{g(x)+2}{f(x)-3}$? Explain. [2 points]

Free-Response Practice – Calculator Required

Near $x = 1$, calculator data indicate $p(x) \to 2.500$, $q(x) \to -1.200$, and $r(x) \to 0.400$.

(a) Estimate $\lim \frac{p(x)^2 - q(x)}{r(x)}$. [2 points]

(b) Explain why the quotient law is valid. [2 points]

(c) If the calculator rounded $r(x)$ to 0.000, what additional investigation would be required? [2 points]